

Course Outcomes: CS 402 Analysis Design of Algorithms

2.4.2.1	Analyze and design algorithms using asymptotic notations and divide and conquer techniques; implement algorithms like heap sort, merge sort, and quick sort efficiently.
2.4.2.2	Apply greedy strategies to solve classical optimization problems such as Huffman coding, minimum spanning trees, and shortest path algorithms
2.4.2.3	Formulate and solve problems using dynamic programming approaches including knapsack, multistage graph, and Floyd-Warshall algorithms.
2.4.2.4	Develop solutions using backtracking and branch & bound techniques for combinatorial problems like the 8-queens, Hamiltonian cycle, and traveling salesman problem.
2.4.2.5	Utilize advanced tree and graph data structures and traversal methods; understand computational complexity concepts including NP-completeness and their implications on problem-solving.

List of Experiments:

SN	Experiment	Course Outcome
1	Draw flowchart, make algorithm then write Pseudocode and finally code to: Find the maximum and minimum marks (Modular Approach), Calculate the average marks, count the number of students scoring above average	CO1
2	Write a program to implement Iterative and Recursive Binary Search.	CO1
3	Write a program for Strassen's Matrix Multiplication.	CO1
4	Write a program for Single Source Shortest Path using Dijkstra's Algorithm.	CO2
5	Write a program for Job Sequencing with Deadlines.	CO2
6	Write a program for Fractional Knapsack Problem.	CO2
7	Write a program to implement Fibonacci Series using Recursion and Dynamic Programming	CO3
8	Write a program to find the Longest Subsequence Common to two given strings.	CO3
9	Write a program to find the Hamiltonian Cycle in a graph and display the cycle path.	CO4
10	Write a program to implement N-Queens Problem using Backtracking.	CO4
11	Write a program for Tree Traversals (Inorder, Preorder, Postorder).	CO5
12	Write a program to implement B-Tree (basic operations).	CO5

Note: Implementation is allowed in any programming language: C/C++/Java/Python etc.